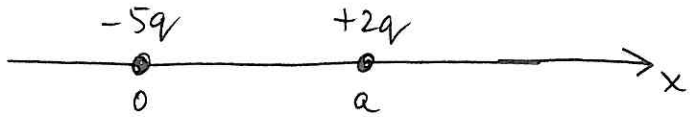
P1

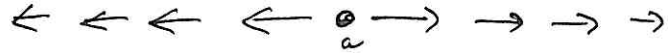
(a)



Electric field of point charges is radial ⇒ the only line where it can be zero is x-axis as elsewhere the total field is a sum of two non-collinear vectors.

On the $\hat{x}$-axis:

$$\vec{E}_{+2q} = \frac{2q}{4\pi\varepsilon_0 (x-a)^2} \cdot \begin{cases} \hat{x}, & x>a \\ -\hat{x}, & x<a \end{cases}$$



$$\vec{E}_{-5q} = \frac{-5q}{4\pi\varepsilon_0 x^2} \cdot \begin{cases} \hat{x}, & x>0 \\ -\hat{x}, & x<0 \end{cases}$$

Thus for $\underline{x<0}$:

$$\frac{5q\hat{x}}{x^2} + \frac{-2q\hat{x}}{(x-a)^2} = 0$$

$$5(x-a)^2 = 2x^2$$

$$3x^2 - 10ax + 5a^2 = 0, \quad x = \frac{5a \pm \sqrt{25a^2 - 15a^2}}{3} = \frac{5 \pm \sqrt{10}}{3} a$$

However $\sqrt{10} < 5 \Rightarrow$ both roots are positive

$\Rightarrow$ no solution for $x<0$.

$\underline{0<x<a}$:

$$\Rightarrow \frac{-5q\hat{x}}{x^2} + \frac{-2q\hat{x}}{(x-a)^2} = 0$$

$$\frac{5}{x^2} + \frac{2}{(x-a)^2} = 0$$

both terms are positive $\Rightarrow$ no solutions for $x \in (0,a)$

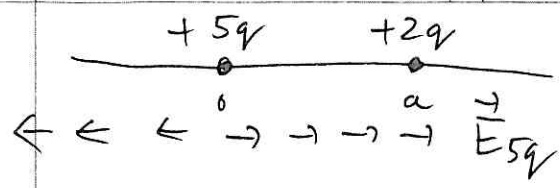
$\underline{x>a}$

$$\frac{-5q\hat{x}}{x^2} + \frac{2q\hat{x}}{(x-a)^2} = 0 \Rightarrow 5(x-a)^2 = 2x^2$$

$$\Rightarrow x = \frac{1}{3}(5 \pm \sqrt{10})a = \begin{cases} \frac{5+\sqrt{10}}{3}a, & x>a \Rightarrow \text{solution} \Rightarrow x = \frac{5+\sqrt{10}}{3}a \\ \frac{5-\sqrt{10}}{3}a, & \text{which is } x<a \Rightarrow \text{no solution} \end{cases}$$

$\leftarrow \leftarrow \leftarrow \leftarrow \mid \rightarrow \rightarrow \rightarrow$ field of $\vec{E}_{2q}$

(b)



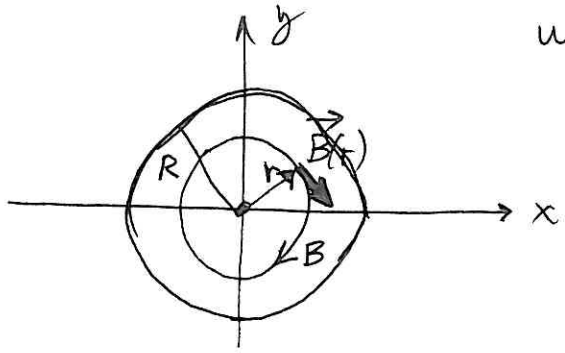
Only region where $\vec{E}_{2q}$ and $\vec{E}_{5q}$ are in opposite directions is $x \in (0, a)$ $\Rightarrow$ look here:

$$\frac{5q\,\hat{x}}{x^2} + \frac{2q(-\hat{x})}{(x-a)^2} = 0 \Rightarrow 5(x-a)^2 = 2x^2$$

Out of two roots, $x = \begin{cases} \frac{5+\sqrt{10}}{3}a, & \text{this is } x > a \\ \frac{5-\sqrt{10}}{3}a, & \text{this is } \underline{x < a} \end{cases}$

we now need to choose one with $0 < x < a$, which means $x = \frac{5-\sqrt{10}}{3}a$.

P2 (a)



uniform $I \Rightarrow J = \frac{I}{\pi R^2}$

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} = \mu_0 \frac{\pi r^2}{\pi R^2} I$$

$$B \cdot 2\pi r = \mu_0 \frac{r^2}{R^2} I = \mu_0 J \pi r^2$$

Magnitude $B = \frac{\mu_0 r}{2\pi R^2} I = \frac{\mu_0 J}{2} r$

Direction $\vec{B}(\vec{r})$ perpendicular to $\vec{r}$,

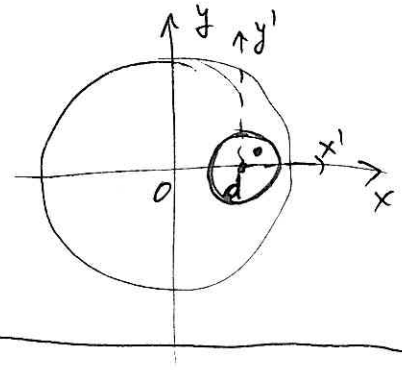
and clockwise $\Rightarrow \hat{B} = \frac{y\hat{x} - x\hat{y}}{r} = \frac{y}{r}\hat{x} - \frac{x}{r}\hat{y}$

$$\Rightarrow \vec{B} = \frac{\mu_0 I}{2\pi R^2}\left(y\hat{x} - x\hat{y}\right) = \frac{1}{2}\mu_0 J\left(y\hat{x} - x\hat{y}\right)$$

Another way to get the same: $\hat{B} = -\hat{\varphi} = (\sin\varphi\,\hat{x} - \cos\varphi\,\hat{y}) = \frac{y}{r}\hat{x} - \frac{x}{r}\hat{y}$.

(b) No current inside cylindrical hole $\Rightarrow$ can use superposition current with density J <u>into the page</u> in the main wire and current with J <u>out of the page</u> in the hole (total inside the hole $= 0$)

$J(\pi R^2 - \pi a^2) = I$



$$\vec{B} = \vec{B}_{main} + \vec{B}_{hole}$$

$$\vec{B}_{main} = \frac{1}{2}\mu_0 J\left(y\hat{x} - x\hat{y}\right)$$

$$\vec{B}_{hole} = \frac{1}{2}\mu_0(-J)\left(y\hat{x} - (x-d)\hat{y}\right)$$

differently $\vec{B}_{hole} = \frac{1}{2}\mu_0(-J)\left(y'\hat{x}' - x'\hat{y}'\right)$ inside radius a

where (x', y') are coordinates centered on the axis of the hole.

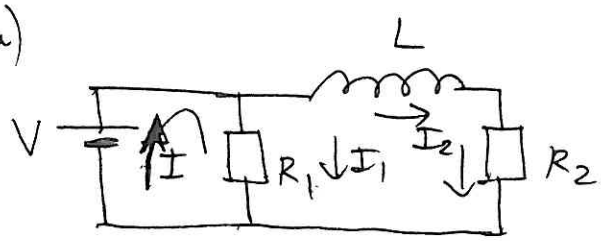
But $\hat{x}' = \hat{x}$, $\hat{y}' = \hat{y}$ and $y' = y$, $x' = x - d$

Thus $\vec{B} = \frac{1}{2}\mu_0 J(-d\hat{y})$

$= -\frac{1}{2}\mu_0 J d\,\hat{y}$ inside the hole.

$\vec{B} = \frac{-\mu_0 d I}{\pi(R^2 - a^2)}\hat{y}$ the field <u>is uniform</u>

P3 (a)



Steady state $\Rightarrow I = \text{const}$

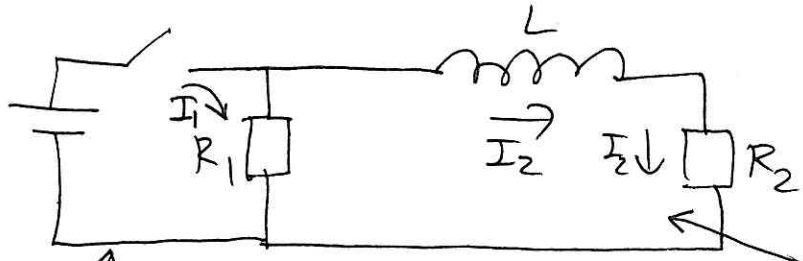
$\Rightarrow$ no back emf due to L.

$$I = I_1 + I_2 = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\left.\begin{aligned} I_1 R_1 &= V \\ I_2 R_2 &= V \end{aligned}\right\}$$

$$\Rightarrow I = \left(\frac{1}{R_1} + \frac{1}{R_2}\right) V = \frac{R_1 + R_2}{R_1 R_2} V, \quad R_{Total} = \frac{R_1 R_2}{R_1 + R_2}$$

(b)



open circuit $I_1 R_1 = 0 \Rightarrow I_1 = 0$

$$-L \frac{dI_2}{dt} = (R_1 + R_2) I_2$$

$$I_2(t) = I_2(0)\, e^{-t\frac{(R_1+R_2)}{L}}$$

But $I_2(0) = \dfrac{V}{R_2}$ = current through inductor L at $t=0$.

$$I_2(t) = \frac{V}{R_2}\, e^{-\frac{t}{L}(R_1+R_2)}$$

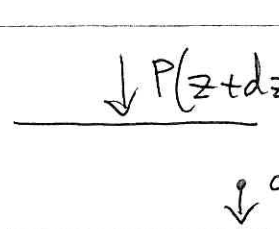
(c) $$W = \int_0^T (I_2(t))^2 (R_1 + R_2)\, dt = (R_1+R_2)\left(\frac{V}{R_2}\right)^2 \int_0^T e^{-\frac{2(R_1+R_2)}{L}t}\, dt$$

$$= \frac{1}{2} L \left(\frac{V}{R_2}\right)^2 \int_0^{\frac{2(R_1+R_2)T}{L}} e^{-s}\, ds = -\frac{1}{2} L \left(\frac{V}{R_2}\right)^2 \left(e^{-\frac{2(R_1+R_2)T}{L}} - 1\right).$$

Note that $W > 0 \Rightarrow$ energy is lost. This energy comes from magnetic field energy stored in the inductor $W \xrightarrow[T\to\infty]{} +\frac{1}{2} L \left(\frac{V}{R_2}\right)^2 = +\frac{1}{2} L I_2$

The total dissipated energy = initial magnetic field energy.

P4



$$\big(P(z) - P(z+dz)\big) A = \rho g A dz$$

$$P(z+dz) = P(z) + \frac{dP}{dz} dz$$

(a)

$$-\frac{dP}{dz} dz A = \rho g A dz$$

$$\frac{dP}{dz} = -\rho g \quad ; \quad \text{density } \rho = m\frac{N}{V} = m\frac{P}{k_B T}$$

$$PV = N k_B T$$

$$\frac{dP}{dz} = -\frac{mg}{k_B T} P$$

(b)

$$P(z) = P(0)\, e^{-\frac{mgz}{k_B T}} \quad \text{where } P(0) = \text{pressure at } z=0.$$

for $T = \text{const}$ (that is, independent of z)

(c)

Since $\rho = \frac{mP}{k_B T} \Rightarrow \rho(z) = \frac{mP(z)}{k_B T} = \frac{mP(0)}{k_B T} e^{-\frac{mgz}{k_B T}}$

$$\Rightarrow \rho(z) = \rho(0)\, e^{-\frac{mgz}{k_B T}}$$

$$\rho(0) = \frac{mP(0)}{k_B T}$$

Density obeys the same exponential law as pressure P.